

TCL CSOT First 14inch 2.8K Inkjet Printing OLED Display

Weiran Cao*, Baixiang Han*, Jinchaun Li*, Yunpeng Luo*, Liying Zhang*, Jingchun Guo*

*Wuhan China Star Optoelectronics Technology Co. Ltd., Wuhan, China

caowr@tcl.com

Abstract

IJP AMOLED, as for the convenient and high-output ability, holds the demission tickets for next generation mass production. Printing process has high requirement on OLED stack-up, which will affect the cavity of OLED device. However, the flatness control inside device is more difficult compared with FMM. TCL CSOT tackles this through process flow optimization, and achieves R-shape printing for the first time. In this paper, we delivered 14-inch IJP high-resolution AMOLED display fabricated on G5.5 factory.

Author Keywords

OLED; Inkjet printing; R-shape printing; Uniformity.

1. Introduction

Display technology has evolved from backlight-dependent Liquid Crystal Display (LCD) to self-emissive and high-contrast Organic Light Emitting Diode (OLED). Currently, the mainstream OLED process is based on high-precision fine metal mask (FMM) evaporation, but challenges such as low material utilization and large-size production confinement still exist and contradict development trends. The industry is expected to usher in a new chapter by new OLED technology. Ink-jetting printing (IJP) OLED, as newly technology, has been developed and iterated for over decades [1-2]. The feasible manufacturing process made it capable on different size of panels, range from smart-size phone to medium-size monitor. Up to now, TCL CSOT is the only manufacturer of inkjet printing OLED technology in the world. For past 12 years, TCL CSOT has been investigating in this field and has realized mass production in 2024.

The structure of AMOLED devices greatly affects their efficiency, unavoidably resulting in excessively high voltage (ELVDD-ELVSS) across the circuit and corresponding high energy consumption. Printed layers are sensitive to the down-side substrate, which means the effect will be amplified inside IJP OLED structure. Specifically, the flatness of film layer beneath the light-emitting layer (EML) in thin-film devices greatly affects the final printed EML, which cannot be directly improved through Vacuum Drying (VCD) manufacturing process. For daily portable notebook, the usage of rounded corner display represents an optimal balance between engineering reliability, user-friendly human-computer interaction, and modern aesthetics, achieved as the consumer electronics industry matures. Like traditional OLED evaporation fabrication process, such as FMM, IJP OLED meet problems during R-shape corner printing as well.

In this paper, we present a 14inch AMOLED LTPS display, which features with a 2880 x 1800 high resolution. Confined by two-type pixel definition layer (PDL, bank), the aperture ratio of our panel can reach larger than 40%. To reduce segment inside stack-up, this paper proposes first time that organic layer (PLNS) fills the gaps inner metal layer (SD) for circuit metal traces during array process flow, thereby to reduce the segment caused by film accumulation. The R-shape printing design was achieved by improving pixel arrangement design and corresponding compensation algorithm.

These all ensure the application of products.

2. Results and Discussion

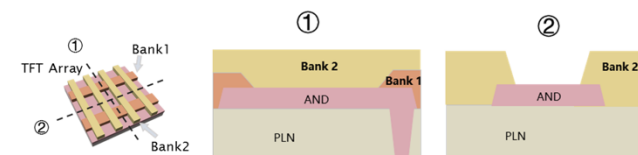


Figure 1. PDL design in IJP OLED.

Essentially, the array substrate and driving system are common to different OLED technologies, while the manufacturing processes of the OLED devices vary among them. Different from traditional OLED, IJP OLED has 2 banks to define the pixel area. Bank1 and Bank2 will be manufactured as PDL, featuring with different material properties and functionalities. Bank1 with hydrophilic property is utilized to separate pixel of the same color that align with vertical side of panel, ensuring the ink material uniformly flowing over the pixels in the later stage. For the layer of Bank2, hydrophobic material is used to separate pixel of different adjacent color, preventing the pixels with different color from mixing to each other. Figure 1 shows the details of pixel bank design.

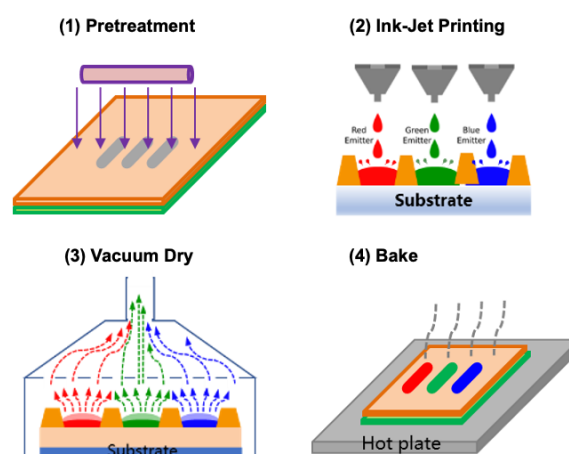


Figure 2. Four steps in IJP OLED process: (1) Pretreatment, (2) Ink-Jet Printing, (3) Vacuum Dry and (4) Bake.

After the preparation of previous process, glass substrate will be transferred to Electro luminance (EL) line to start the OLED device fabrication. The main IJP technology will be employed to fabricate OLED device layers of HIL, HTL and EML, comprising four key steps: pre-treatment, ink-jet printing, vacuum dry and bake, as illustrated in Figure 2. The pre-treatment procedure aims at enhancing the anode surface energy and lowering OLED device driving voltage. During the process of inkjet printing, precise control over droplet position and ink volume uniformity is achieved, permitting high patterning accuracy. Once the pixels are filled with the desired amount of ink, the substrate is transferred to a vacuum chamber for drying. This vacuum dry process helps

suppress the coffee ring effect and ensures uniform film formation by promoting homogeneous solvent evaporation. Finally, the substrate undergoes a bake step to complete solvent removal, thereby optimizing the film's morphological and electrical properties.

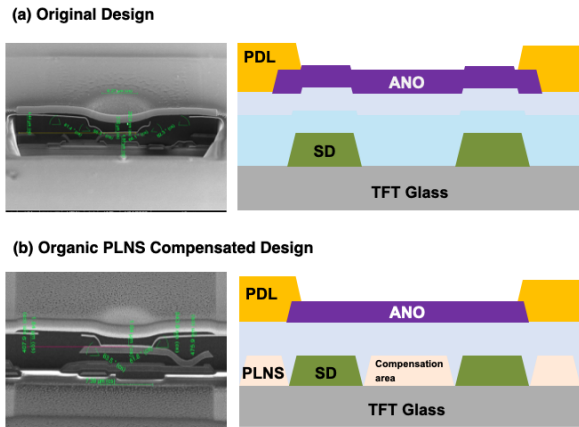


Figure 3. Cross-sectional comparison between (a) original design and PLNS-applied Design.

Since IJP process is highly sensitive to the morphology beneath printed layers, we were trying to decrease segment through array process. Due to the differences existed inside the metal stacking structure between capacitor and trace regions, low-temperature poly-silicon (LTPS) circuits inevitably exhibit significant segment differences along the vertical data line direction. Figure 3(a) shows the cross-sectional view of this original design. These minute fluctuations bring disadvantages on subsequent IJP films formation control. Organic PLN layer working as compensation layer, same height as SD layer, are applied to fill the gap. Combined with exchanged exposure model and bank process flow, the impact of segment is further reduced and related design is shown in Figure 3(b). Compared with original design, PLN thickness in compensated structure is decreased 50% and the depth of anode hole was further narrowed. The benefits finally realized through improved efficiency and pixel uniformity. For the benefits on green and blue, the efficiency increases for 20% and 40%, respectively.

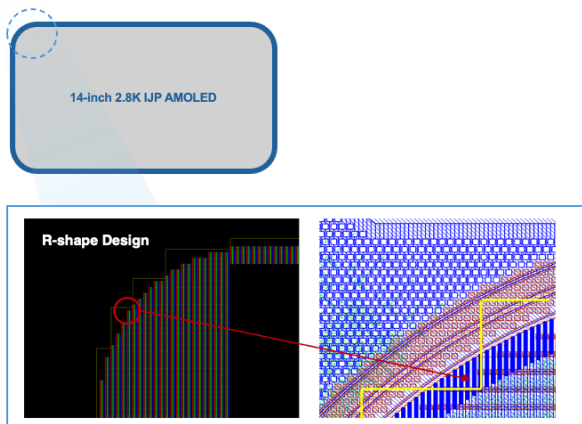


Figure 4. R-shape design in 14inch IJP OLED.

The R-corner shape design on the panel is one of the product specifications driven by marketing requirements. For IJP OLED, however, there is no prior experience in R-corner printing. Due to the limitations in current software capabilities, the R-corner recipe is manually edited and designed with rectangular blocks, leading to incomplete printing of certain pixel columns. As a result, during panel illumination, mura issue appears in the R-corner region due to the inconsistencies in film thickness compared to the central area.

To break through the challenge, TCL CSOT has developed the unique shape printing technology with the equipment vendor to fulfill the product spec. The software has been developed with following steps and realize through: automatically reading of printing area patterns from CAD drawings, corresponding coordinate information identification, setting the cell pattern design and creating the recipe to realize the target patterns.



Figure 5. Performance of 14inch 2.8K IJP AMOLED Display.

High resolution based on IJP technology was then further evaluated on LTPS AMOLED backplanes. Figure 5 shows the real performance of 14inch 2.8K IJP AMOLED display. Attributed to the unique strip pixel design, the display shows no color clarity and nature frame transition. With further improvement of printing accuracy, the limitation of PPI will be broken in the very next future. The successful product fabrication demonstrates that G5.5 line has indeed the capability to achieve high resolution display mass production.

The detailed specification of 14inch 2.8K panel is listed inside Table 1. Higher refresh rate and TED design based on oxide internal circuit [3] are still undergoing since this type of demand from market is increasing rapidly.

Table 1. The Specification of 14inch Panel

Panel Specification	
Panel Type	AMOLED
Panel Size	14inch
Backplane	LTPS TFT
Resolution	2880 x RGB x 1800
Pixel Pitch	240 PPI (Strip Real RGB)
Refresh Rate	60~120 Hz

3. Conclusion

With a pixel pitch of 240, the 14inch panel we discussed here demonstrated a high-quality performance and varied the feasibility of mass production for IJP Process. The segment caused efficiency loss has been improved through array process optimization and additional PLN mask, segment inside the device stack-up now decreased smaller than 100nm. Besides, R-shape printing was achieved for the first time through overcoming process and printing difficulties, the 14inch display based on that shows better visual performance. More importantly, TCL CSOT based on IJP OLED are trying to deliver high quality and various size products in an environmental-friendly and low-carbon way, and the further mass production rely on IJP technology and oxide backplane will also be delivered based on our G8.6 factory.

4. References

1. Suzuki M, Fukagawa H, Nakajima Y, Tsuzuki T, Takei T, Yamamoto T, et al. A 5.8-in. phosphorescent color AMOLED display fabricated by ink-jet printing on plastic substrate. *Journal of the Society for Information Display*. 2009 Dec;17(12):1037-42.
2. Kant C, Shukla A, McGregor SK, Lo SC, Namdas EB, Katiyar M. Large area inkjet-printed OLED fabrication with solution-processed TADF ink. *Nature Communications*. 2023 Nov 9;14(1):7220.
3. Cao W, Li J, Li XL, Liu Y, Hsu YJ, Wu YC, et al. 33-2: Invited Paper: Flexible OLED Displays with Ink-Jet Printing Technology. *InSID Symposium Digest of Technical Papers* 2022 Jun (Vol. 53, No. 1, pp. 395-397)